

```
In [1]: import pandas as pd
```

```
In [2]: sql = pd.read_csv(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Classwork\Nov 2025\18th Nov - Data Analysis
```

```
In [3]: sql
```

```
Out[3]:
```

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	Carry/
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	...	
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner	...	
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	...	
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner	...	
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner	...	
...	...	...	...	...	...	...	...	...	...	...	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Male	26	Single	...	
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	Male	26	Single	...	
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	Male	26	Single	...	
12682	Work	Alone	Snowy	30	7AM	Bar	1d	Male	26	Single	...	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Male	26	Single	...	

12684 rows × 27 columns



```
In [4]: sql[['weather', 'temperature']]
```

```
Out[4]:
```

	weather	temperature
--	---------	-------------

0	Sunny	55
1	Sunny	80
2	Sunny	80
3	Sunny	80
4	Sunny	80
...	...	...
12679	Rainy	55
12680	Rainy	55
12681	Snowy	30
12682	Snowy	30
12683	Sunny	80

12684 rows × 2 columns

```
In [5]: sql.head(10)
```

Out[5]:

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	CarryAway
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	...	NaN
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner	...	NaN
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	...	NaN
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner	...	NaN
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner	...	NaN
5	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	2h	Female	21	Unmarried partner	...	NaN
6	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	1d	Female	21	Unmarried partner	...	NaN
7	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	2h	Female	21	Unmarried partner	...	NaN
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	...	NaN
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Female	21	Unmarried partner	...	NaN

10 rows × 27 columns



In [6]: `sql['passanger'].unique`

```
Out[6]: <bound method Series.unique of 0      Alone
1      Friend(s)
2      Friend(s)
3      Friend(s)
4      Friend(s)
...
12679    Partner
12680      Alone
12681      Alone
12682      Alone
12683      Alone
Name: passanger, Length: 12684, dtype: object>
```

```
In [10]: sql[sql['destination'] == 'Home']
```

Out[10]:

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	Carry/
13	Home	Alone	Sunny	55	6PM	Bar	1d	Female	21	Unmarried partner	...	
14	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	Female	21	Unmarried partner	...	
15	Home	Alone	Sunny	80	6PM	Coffee House	2h	Female	21	Unmarried partner	...	
35	Home	Alone	Sunny	55	6PM	Bar	1d	Male	21	Single	...	
36	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	Male	21	Single	...	
...	...	...	...	...	...	...	...	...	...	...	...	
12675	Home	Alone	Snowy	30	10PM	Coffee House	2h	Male	26	Single	...	
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	Male	26	Single	...	
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1d	Male	26	Single	...	
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h	Male	26	Single	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	Male	26	Single	...	

3237 rows × 27 columns



In [11]:

sql.sort_values('coupon')

Out[11]:

	destination	passanger	weather	temperature	time		coupon	expiration	gender	age	maritalStatus	...	Car
11702	Home	Partner	Sunny	30	10PM		Bar	2h	Female	50plus	Married partner	...	
9930	No Urgent Place	Alone	Snowy	30	2PM		Bar	1d	Female	21	Single	...	
10632	Home	Alone	Rainy	55	6PM		Bar	1d	Male	21	Single	...	
7997	No Urgent Place	Friend(s)	Rainy	55	10PM		Bar	2h	Male	26	Unmarried partner	...	
11166	Work	Alone	Snowy	30	7AM		Bar	1d	Female	41	Married partner	...	
...	...	...	...	...	...		...	...	...	...	...	...	
10476	Home	Alone	Sunny	80	6PM	Restaurant(<20)		1d	Female	31	Unmarried partner	...	
5447	Home	Alone	Sunny	80	10PM	Restaurant(<20)		2h	Female	50plus	Single	...	
10478	Home	Alone	Snowy	30	10PM	Restaurant(<20)		2h	Female	31	Unmarried partner	...	
5440	No Urgent Place	Alone	Sunny	80	2PM	Restaurant(<20)		2h	Female	50plus	Single	...	
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)		1d	Female	21	Unmarried partner	...	

12684 rows × 27 columns



In [12]:

sql['destination']

```
Out[12]: 0      No Urgent Place
        1      No Urgent Place
        2      No Urgent Place
        3      No Urgent Place
        4      No Urgent Place
        ...
        12679      Home
        12680      Work
        12681      Work
        12682      Work
        12683      Work
        Name: destination, Length: 12684, dtype: object
```

```
In [16]: sql.rename(columns={'destination':'Destination'},inplace = True)
        sql['Destination']
```

```
Out[16]: 0      No Urgent Place
        1      No Urgent Place
        2      No Urgent Place
        3      No Urgent Place
        4      No Urgent Place
        ...
        12679      Home
        12680      Work
        12681      Work
        12682      Work
        12683      Work
        Name: Destination, Length: 12684, dtype: object
```

```
In [24]: sql.groupby('occupation').size()
```

```
Out[24]: occupation
Architecture & Engineering      175
Arts Design Entertainment Sports & Media  629
Building & Grounds Cleaning & Maintenance  44
Business & Financial            544
Community & Social Services      241
Computer & Mathematical         1408
Construction & Extraction        154
Education&Training&Library      943
Farming Fishing & Forestry       43
Food Preparation & Serving Related  298
Healthcare Practitioners & Technical  244
Healthcare Support              242
Installation Maintenance & Repair  133
Legal                           219
Life Physical Social Science     170
Management                      838
Office & Administrative Support    639
Personal Care & Service           175
Production Occupations          110
Protective Service              175
Retired                         495
Sales & Related                  1093
Student                         1584
Transportation & Material Moving   218
Unemployed                     1870
dtype: int64
```

```
In [28]: sql.groupby('occupation').size().to_frame('Count')
```

Out[28]:

	Count
occupation	
Architecture & Engineering	175
Arts Design Entertainment Sports & Media	629
Building & Grounds Cleaning & Maintenance	44
Business & Financial	544
Community & Social Services	241
Computer & Mathematical	1408
Construction & Extraction	154
Education&Training&Library	943
Farming Fishing & Forestry	43
Food Preparation & Serving Related	298
Healthcare Practitioners & Technical	244
Healthcare Support	242
Installation Maintenance & Repair	133
Legal	219
Life Physical Social Science	170
Management	838
Office & Administrative Support	639
Personal Care & Service	175
Production Occupations	110
Protective Service	175
Retired	495

	Count
occupation	
Sales & Related	1093
Student	1584
Transportation & Material Moving	218
Unemployed	1870

```
In [31]: #select occupation from dataset_1 group by occupation;    -SQL
sql.groupby('occupation').size().to_frame('Count').reset_index()
```

Out[31]:

	occupation	Count
0	Architecture & Engineering	175
1	Arts Design Entertainment Sports & Media	629
2	Building & Grounds Cleaning & Maintenance	44
3	Business & Financial	544
4	Community & Social Services	241
5	Computer & Mathematical	1408
6	Construction & Extraction	154
7	Education&Training&Library	943
8	Farming Fishing & Forestry	43
9	Food Preparation & Serving Related	298
10	Healthcare Practitioners & Technical	244
11	Healthcare Support	242
12	Installation Maintenance & Repair	133
13	Legal	219
14	Life Physical Social Science	170
15	Management	838
16	Office & Administrative Support	639
17	Personal Care & Service	175
18	Production Occupations	110
19	Protective Service	175
20	Retired	495
21	Sales & Related	1093

	occupation	Count
22	Student	1584
23	Transportation & Material Moving	218
24	Unemployed	1870

```
In [34]: sql.groupby('weather')['temperature'].mean()
```

```
Out[34]: weather
Rainy    55.000000
Snowy    30.000000
Sunny    68.946271
Name: temperature, dtype: float64
```

```
In [35]: #select weather, avg(temperature) as avg_temp from dataset_1 group by weather;
sql.groupby('weather')['temperature'].mean().to_frame('avg_temp').reset_index()
```

```
Out[35]:   weather  avg_temp
0    Rainy  55.000000
1    Snowy  30.000000
2    Sunny  68.946271
```

```
In [37]: #select weather, count(temperature) as count_temp from dataset_1 group by weather;
sql.groupby('weather')['temperature'].size().to_frame('count_temp').reset_index()
```

```
Out[37]:   weather  count_temp
0    Rainy           1210
1    Snowy           1405
2    Sunny           1069
```

```
In [38]: #select weather, count(distinct temperature) as count_distinct_temp from dataset_1 group by weather;
sql.groupby('weather')['temperature'].nunique().to_frame('count_distinct_temp').reset_index()
```

Out[38]:

	weather	count_distinct_temp
0	Rainy	1
1	Snowy	1
2	Sunny	3

```
In [39]: #select weather, sum(temperature) as sum_temp from dataset_1 group by weather;
sql.groupby('weather')['temperature'].sum().to_frame('sum_temp').reset_index()
```

Out[39]:

	weather	sum_temp
0	Rainy	66550
1	Snowy	42150
2	Sunny	694220

```
In [40]: #select weather, min(temperature) as min_temp from dataset_1 group by weather;
sql.groupby('weather')['temperature'].min().to_frame('min_temp').reset_index()
```

Out[40]:

	weather	min_temp
0	Rainy	55
1	Snowy	30
2	Sunny	30

```
In [41]: #select weather, max(temperature) as max_temp from dataset_1 group by weather;
sql.groupby('weather')['temperature'].max().to_frame('max_temp').reset_index()
```

```
Out[41]:
```

	weather	max_temp
0	Rainy	55
1	Snowy	30
2	Sunny	80

```
In [46]: #select occupation from dataset_1 group by occupation having occupation = 'Student';
sql.groupby('occupation').filter(lambda x : x['occupation'].iloc[0] == 'Student').groupby('occupation').size()
```

```
Out[46]: occupation
Student    1584
dtype: int64
```

```
In [91]: #Read table_to_union
sql1 = pd.read_csv(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Classwork\Nov 2025\18th Nov - Data Analysis\table_to_union.csv')
sql1
```

```
Out[91]:
```

	destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	CarryAway
0	UNION	UNION	UNION	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	...	NaN

1 rows × 27 columns



```
In [93]: #SELECT DISTINCT destination FROM(SELECT * FROM dataset_1 UNION SELECT * FROM table_to_union)
pd.concat([sql,sql1])['Destination'].drop_duplicates()
```

```
Out[93]: 0    No Urgent Place
13              Home
16              Work
0              NaN
Name: Destination, dtype: object
```

```
In [90]: #Read table_to_join
sql2 = pd.read_csv(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Classwork\Nov 2025\18th Nov - Data Analysis\table_to_join.csv')
sql2
```

Out[90]:

	time	part_of_day
0	2PM	Afternoon
1	10AM	Morning
2	6PM	Evening
3	7AM	Morning
4	10PM	Night

```
In [89]: #SELECT a.destination,a.time,b.part_of_day FROM dataset_1 a INNER JOIN table_to_join b ON a.time=b.time
pd.merge(sql, sql2[['time', 'part_of_day']], on='time', how='inner')[['Destination', 'time', 'part_of_day']]
```

Out[89]:

	Destination	time	part_of_day
0	No Urgent Place	2PM	Afternoon
1	No Urgent Place	10AM	Morning
2	No Urgent Place	10AM	Morning
3	No Urgent Place	2PM	Afternoon
4	No Urgent Place	2PM	Afternoon
...	...	...	...
12679	Home	6PM	Evening
12680	Work	7AM	Morning
12681	Work	7AM	Morning
12682	Work	7AM	Morning
12683	Work	7AM	Morning

12684 rows × 3 columns

```
In [88]: #select destination, passanger from (select * from dataset_1 where passanger = 'Alone');
#select destination, passanger from dataset_1 where passanger = 'Alone';
```

```
sql[sql['passanger']=='Alone'][['Destination','passanger']]
```

Out[88]:

	Destination	passanger
0	No Urgent Place	Alone
13	Home	Alone
14	Home	Alone
15	Home	Alone
16	Work	Alone
...	...	...
12676	Home	Alone
12680	Work	Alone
12681	Work	Alone
12682	Work	Alone
12683	Work	Alone

7305 rows × 2 columns

```
In [67]: #select * from dataset_1 where weather Like 'Sun%';  
sql[sql['weather'].str.startswith('Sun')]
```

Out[67]:

	Destination	passanger	weather	temperature	time	coupon	expiration	gender	age	maritalStatus	...	Carry
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female	21	Unmarried partner	...	
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female	21	Unmarried partner	...	
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female	21	Unmarried partner	...	
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female	21	Unmarried partner	...	
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female	21	Unmarried partner	...	
...	...	...	...	...	...	...	...	...	...	...	...	
12673	Home	Alone	Sunny	30	6PM	Carry out & Take away	1d	Male	26	Single	...	
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	Male	26	Single	...	
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1d	Male	26	Single	...	
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h	Male	26	Single	...	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	Male	26	Single	...	

10069 rows × 27 columns



```
In [80]: #select distinct temperature from dataset_1 where temperature between 29 and 75;
sql[(sql['temperature'] >= 29) & (sql['temperature'] <= 75)]['temperature'].unique()
```

Out[80]: array([55, 30])

```
In [82]: sql[sql['occupation'].isin(['Sales & Related','Management'])][['occupation']]
```

Out[82]:

	occupation
193	Sales & Related
194	Sales & Related
195	Sales & Related
196	Sales & Related
197	Sales & Related
...	...
12679	Sales & Related
12680	Sales & Related
12681	Sales & Related
12682	Sales & Related
12683	Sales & Related

1931 rows × 1 columns

In []: